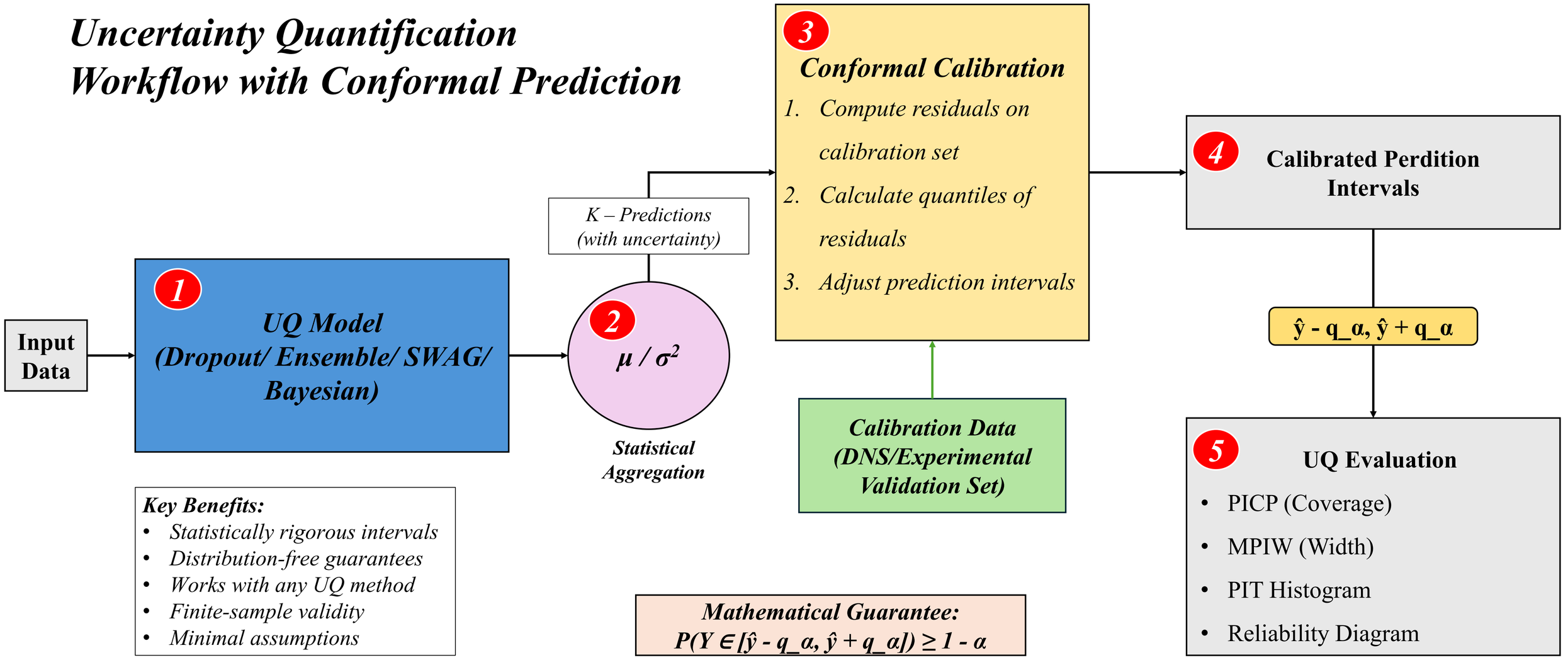

Uncertainty Quantification
Workflow with Conformal Prediction
Input Data
1
UQ Model (Dropout/ Ensemble/ SWAG/ Bayesian)
2
μ / σ²
Statistical Aggregation
K – Predictions (with uncertainty)
3
Conformal Calibration
1. Compute residuals on calibration set
2. Calculate quantiles of residuals
3. Adjust prediction intervals
Calibration Data (DNS/Experimental Validation Set)
4
Calibrated Perdition Intervals
ŷ - q_α, ŷ + q_α
5
UQ Evaluation
• PICP (Coverage)
• MPIW (Width)
• PIT Histogram
• Reliability Diagram
Key Benefits:
• Statistically rigorous intervals
• Distribution-free guarantees
• Works with any UQ method
• Finite-sample validity
• Minimal assumptions
Mathematical Guarantee:
P(Y ∈ [ŷ - q_α, ŷ + q_α]) ≥ 1 - α